

Mattéo Clémot, Ph.D. student

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🔗 MClemot

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Education

- 2023 – 2026  **Ph.D. candidate, Université Lyon 1**
Thesis title: *Dimension reduction under topological constraints*.
Supervised by Julien Tierny (LIP6, Sorbonne Université) & Julie Digne (LIRIS, Lyon 1).
- 2019 – 2023  **M.Sc. in fundamental computer science, École Normale Supérieure de Lyon**
- 2018 – 2019  **Engineering school, École Nationale des Ponts et Chaussées**
- 2016 – 2018  **Preparatory classes, Lycée Condorcet, Paris**
Two-year undergraduate intensive course in mathematics and physics, with computer science option.

Research internships

- Feb.–Apr. 2023  **The University of Edinburgh**, School of Informatics, Institute of Perception, Action and Behaviour
Vector-field design on implicit surfaces. Supervised by Amir Vaxman.
- Oct.–Dec. 2022  **LIP**, Team Ockham, Lyon
Solving ill-posed inverse problems involving partial differential equations with neural networks. Supervised by Elisa Riccietti & Stefania Bellavia.
- Feb.–July 2022  **LIRIS**, Team Origami, Lyon
Learning the topology of shapes given by point clouds. Supervised by Julie Digne.
- Apr.–July 2021  **GIPSA-lab**, Team GAIA, Grenoble
Surface reconstruction with Delaunay complexes. Supervised by Dominique Attali.
- June–July 2020  **INRIA Saclay / École polytechnique**, Team Tropical
Links between tropical linear programming, mean payoff games and parity games. Supervised by Stéphane Gaubert & Xavier Allamigeon.

Publications

- 1 **M. Clémot**, J. Digne, and J. Tierny, “Topological autoencoders++: Fast and accurate cycle-aware dimensionality reduction,” *IEEE TVCG*, 2026.  DOI: 10.1109/TVCG.2025.3644671.
- 2 D. Attali, **M. Clémot**, B. B. Dornelas, and A. Lieutier, “When alpha-complexes collapse onto codimension-1 submanifolds,” in *SoCG*, 2025.  DOI: 10.4230/LIPIcs.SocG.2025.11.
- 3 **M. Clémot**, J. Digne, and J. Tierny, *Delaunay–Rips filtration: a study and an algorithm*, preprint, 2025. arXiv: 2512.17382 [cs.CG].  URL: <https://arxiv.org/abs/2512.17382>.
- 4 **M. Clémot** and J. Digne, “Neural skeleton: Implicit neural representation away from the surface,” *Computers & Graphics*, 2023, (SMI best paper award).  DOI: 10.1016/j.cag.2023.06.012.

Talks

- Jul. 2023  **SMI 2023**
Neural skeleton: implicit neural representation away from the surface.
- June 2025  **Journée VISU 2025**
Réduction de dimension sous contraintes topologiques.
- Nov. 2025  **JFIG 2025**
Réduction de dimension sous contraintes topologiques.

Technical skills

- Languages  C++, Python, OCaml, GLSL
- Libraries  Deep Learning (PyTorch & LibTorch), Geometry/Topology (TTK, CGAL, Gudhi), Python bindings (nanobind)...
- Misc. tools  Git, CMake, SQL, $\LaTeX$...
- Experience  Contributor to the Topology ToolKit (TTK) open-source library.

Teaching experience

- 2023 – 2026  **Courses and tutorials**, ISFA (Institut de Science Financière et d'Assurances), Lyon.
 - L3 Actuarial science – C++ (course and tutorial)
 - L3 Actuarial science – Probability (tutorial)
 - M1 Actuarial science – Databases & Microsoft Access (course and tutorial)
 - M1 Econometrics & statistics – Advanced probability (tutorial)
- 2020 – 2023  **Mathematics examiner in preparatory classes**, Lycée La Martinière Monplaisir, Lyon.
Weekly oral exams (called *Colles*) in mathematics in first year of scientific preparatory classes (*Classes Préparatoires aux Grandes Écoles*: two-year undergraduate intensive course in mathematics and physics).
- 2019 – 2020  **Volunteer private tutoring**, ENSeigner, Lyon.
Volunteer private tutoring in mathematics and physics.
- 2018 – 2019  **Private mathematics tutoring**, MyMentor, Paris.
Private tutoring in mathematics.

Miscellaneous

Language certification

- English  **C1 Advanced** (Cambridge English Advanced), obtained in 2022.

Interests

- Sports  Climbing, hiking, trail running...
- Music  Drums and guitar player
- Misc.  Contributor to the french Wikipedia